

Tiki: Scaling dynamically with Google Cloud to serve customers on the busiest days

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ABOUT TIKI

Tiki is able to scale to improve security, high availability to be competitive in the ecommerce industry by migrating its app from on-premise to Cloud.



About Tiki

Tiki is a commerce ecosystem with more than 10 million products across a range of categories and an end-to-end supply chain. According to Nielsen survey, Tiki is the most trusted ecommerce platform in Vietnam. Founded in 2010, Tiki has more than 4,000 employees and is one of the fastest-growing technology companies in the country.

Industries: Technology

Google Cloud results

- Migrates to Google Cloud in just 24 days; one of the fastest migrations to date in Asia Pacific
- Increases primary APIs performance by 30%, from 80ms to 55ms
- Reduces engineering time to allocate new servers from a few days to a few minutes
- Easily scales out the system to seamlessly handle big campaigns when traffic

Location: Vietnam

Helps
98% of
time c
traffic

Cloud Security Command Center

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://cloud.google.com/storage/>), Cloud

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Security Command Center

(<https://cloud.google.com/security-command-center/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud SQL (<https://cloud.google.com/sql/>)

can increase
by 10 times

- Replaces
on-premises
systems in
24 days,
including 12
Kubernetes
clusters,
200+
services,
and 30 TB+
of data

With a name standing for "search and save" in the Vietnamese language, Tiki is a commerce ecosystem in Vietnam with more than 10 million products in its catalog and an end-to-end supply chain. The company has been seeing annual triple-digit growth since it launched in 2010, thanks to a growing middle class in Vietnam (<https://www.worldbank.org/en/country/vietnam/overview>)

"A unique aspect of Vietnam is that this is a market where you see a huge young population that are just beginning to have more spending power. This is where consumption spending will take off, and we are now at the beginning of a long runway, with many years of growth to come," says Tridiv Vasavada, SVP of Technology at Tiki. In order to remain competitive in the ecommerce industry in the country, Tiki turned to Google Cloud (<https://cloud.google.com/>).

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—Hoa Nguyen Manh,
Director of Architecture,
Tiki

Promising speed and convenience

To stand out from the crowd, Tiki differentiates itself with a commitment to authentic products and a fast delivery service. Its TikiNOW two-hour delivery service, for example, is a first-of-its-kind service in Southeast Asia, which allows customers to receive products within two hours of placing an order. "We are now moving into a hyperlocal delivery phase of ecommerce, where consumers are used to ordering things only when they want it, and expect it to be delivered within a short period of time," says Tridiv.

"In a flash sale event, we receive a few thousand orders in one minute, so it's essential that our systems can activate our warehousing workflows within seconds. Especially because we offer two-hour

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*—Tridiv Vasavada, SVP of
Technology, Tiki*

Running Singles' Day sales in the cloud

A key motivator that led Tiki to deploy Google Cloud was its Singles' Day, or 11.11 Legendary Sale. During the recent 11.11 event, it experienced the highest number of orders in a 48-hour time period, where its systems tripled in line availability for the first time.

"In a flash sale event, we receive a few thousand orders in one minute, so it's essential that our systems can activate our warehousing workflows within seconds. Especially because we offer two-hour delivery for many of our products. Thanks to Google Cloud, we're now able to do this," says Tridiv.

In the past, Tiki had experienced technical limitations that affected its customers during peak traffic, such as during flash sales. "We had to constantly prepare our systems to scale and do it manually, since many of our campaigns are

promotions that run only for short periods of time. For example, we would open a campaign at 9 AM and see a 10X spike for that five minutes. It was certainly not easy managing such big spikes throughout the day."

"If we'd surpassed our previous peak sales traffic, we would have certainly struggled with our on-premises arrangement. But what happened instead was that we delivered 98% of orders on time despite the massive spike in traffic because Google Cloud enabled us to scale dynamically," he adds, saying that such an achievement was unparalleled considering the thousands of orders that come in every minute from all around the country.

"We had a hosted infrastructure before, and it took a lot of manual effort from our engineers to maintain our system. Google Cloud lets us implement the

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—*Trung Truong Minh,*
Director of Infrastructure,
Tiki

Tiki has always faced a challenge when load balancing the different kinds of traffic due to its large database. As such, it tends to overscale to be on the safe side. "There has always been a lot of wastage, but all this goes away with Google Cloud," says Trung Truong Minh, Director of Infrastructure at Tiki. "We had a hosted infrastructure before, and it took a lot of manual effort from our engineers to maintain our system. Google Cloud lets us implement the infrastructure as code that is testable and scalable. The new pipeline on Google Cloud is much more stable."

An easy deployment process

Despite having dozens of separate databases, 200 services, and four different languages on its platform, Tiki managed to complete its migration

process in just 24 days, one of the fastest migrations to date in Asia Pacific. "The tools are very simple and usable. It takes the complexity away from users by managing some of the configuration so that users can quickly move large numbers of services at once during the migration process," says Tridiv. Engineers from different backgrounds can quickly understand the programs and tools which results in a more efficient workflow.

He adds that the simplicity of Google Cloud technologies, especially around container management platforms such as Compute Engine (/compute) and Cloud SQL (/sql) also made the transition process from a legacy platform to cloud easy and fast. "Now we have about 70% of our system already in the cloud, 30% of which are warehousing related. We will be moving the rest of the backend systems fully to cloud in the next few months."

Tiki has been using Google Kubernetes Engine (/kubernetes-engine) (GKE) for 200 services and plans to migrate its legacy systems to GKE as well so it can further benefit from the auto-scaling capabilities.



A safer and more secure platform for shoppers

With so many transactions taking place on the platform every second, protecting customer data and ensuring data privacy is paramount. Tridiv says, "We believe that privacy and security goes hand in hand, so by virtue of being in the cloud, you get a high level of encryption in traffic which protects us from misuse of the platform. This is something that we did not have prior to moving to Google Cloud."

Tiki also uses [BigQuery](#) (/bigquery) for data warehousing and [Security Command Center](#) (/security-command-center) to scan traffic across the platform so it has an overview of the types of data that users access. It also uses [TensorFlow](https://www.tensorflow.org/) (https://www.tensorflow.org/) to train and manage models for fraud detection. "Vietnam is a low ASP (average selling price) market, whereby there is high traffic but the value of actual transactions are low. This leads to more fraud problems, and we hope to be able to iron those out," says Trung.

"We will soon be implementing patterns to detect fraud on customer accounts, identify counterfeit

products, fake sellers, and customers who are acting as resellers. We've also seen some cases of coupon codes being used inappropriately and plan to prevent such issues from happening in the future," he adds.



Charting the future of Tiki with Google Cloud

With all the consumer and seller systems currently migrated to the cloud, Tridiv says, "We've now survived the shopping peaks, but as next steps, we're going to look at ways we can optimize better and use the containerization VMs more efficiently."

Tiki will also be focusing more on machine learning, to better forecast demand. "We are already using TensorFlow to train models to detect duplicates in our catalog, but we will be doing more on the AI front in the near future."

